

Anthony Bynum

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Education

Harvard University

Cambridge, MA

Bachelor of Science: Mechanical Engineering with Computer Science secondary. GPA: 3.6

May 2028

Relevant Coursework: Computer-Aided Machine Design, Intro to EE, Mechanics of Solids, Fluid Mechanics and Transport Processes, Thermodynamics, Mechanical Systems, Digital Fabrication, and Humanitarian Design Projects.

Engineering Experience

General Dynamics Mission Systems

Greensboro, NC

Engineering Intern

May 2026 – present

- Designed mechanical assemblies in SolidWorks and architected Python-Blender automation pipeline via Claude MCP for mission scenario visualization.
- Executed trade studies and integration for an undersea demonstration, compiling COTS vehicle options and BOM.

Harvard School of Engineering and Applied Sciences

Cambridge, MA

Teaching Assistant for ES51: Computer-Aided Machine Design

Feb 2025 – present

- Introduced 100+ students to topics such as CAD, 3D printing, GD&T, technical drawings, and CNC machining.
- Facilitated hands-on learning in weekly labs, hosted office hours, and graded homework, design notebooks, and projects.

Mech Tech

Sep 2026 – present

- Perform maintenance and repairs on 3D printers and install digital readout (DRO) systems on manual lathes, while supporting broader shop operations, occasionally assisting with design and fabrication tasks.

Engineering Projects and Clubs

Harvard Rocket Propulsion Group

Fluids and Integration Team Lead

Sep 2024 – present

- Fabricated components for a high-power liquid-bipropellant rocket, including lathe-turning a propellant piston and water jet cutting bulkheads, brackets, and thrust chamber flange plates to engineering tolerances.
- Conducted test fire of the bipropellant engine, performing propellant loading, tank pressurization, and ignition loading.
- Co-designed a liquid rocket injector in SolidWorks, incorporating CFD analysis results to enhance combustion efficiency.

Engineers Without Borders

Kenya Project

Sep 2025 – May 2026

- Used Revit to design and model water kiosks for EWB's Kenya project, improving clean water access in remote villages.

Engineering Design Projects

Sep 2024 – present

- SCARA Robot Arm:** Designing and building a 3-DOF SCARA robotic arm with a single-actuated gripper independently, integrating mechanical, electrical, and software components for precise automated motion.
- Competition Robot** (Spring 2025): Spearheaded design and prototyping of competition robot; achieved top 3 finish; machined structural components via CNC milling, including silicone mold tooling, within strict size and weight constraints.
- Bionic Hand** (Spring 2024): Led circuit design and troubleshooting for myoelectric bionic hand; implemented analog filtering, amplification, and Arduino gesture recognition using Electromyography (EMG) signals.

Research Work

MIT Department of Aeronautics and Astronautics

Sep 2025 – Jan 2026

- Conducting wind-tunnel experiments and aerodynamic analyses to optimize passive dust-mitigation in off-grid solar systems.
- Implementing API calls, completing the user registration system, and advancing the frontend for the Ashesi Solar Monitoring Network (ASMONET) to support expanded solar and weather data collection and analysis in Ghana.

Skills

Engineering: SolidWorks (CSWA-certified), Revit, Onshape, OpenRocket; GD&T, CFD

Manufacturing: CNC mill, lathe, waterjet cutting, 3D printing, laser cutter, soldering

Programming: Python, MATLAB, C++, HTML, CSS, PHP

Languages: English (native), Arabic (advanced), Spanish (intermediate)